

COLDPOD · DESIGN PRECIS

ColdPod design precis

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DOCUMENT CONTROL

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REVISION HISTORY

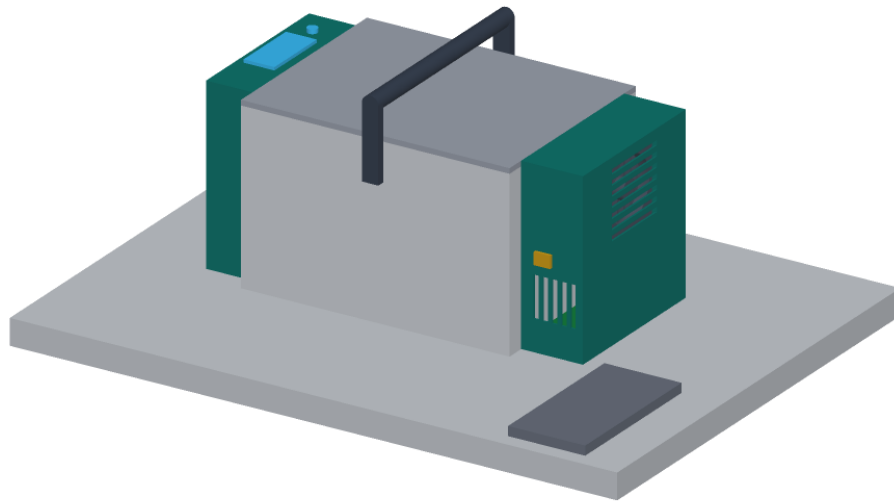
Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; budget \$310, R15 met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Lid cold plate, liner cut-out at 3 °C, thinner printed parts and buck-boost driver; numbers from CPD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (CPD-DDR-001); numbers replaced by CPD-CAL-001; evaporator can, loop thermosiphon and TEC1-12703 class sized; smooth DC drive
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod design precis

ColdPod is a carry case about the size of a lunch cooler (366 x 212 x 207 mm) that holds 1.36 L of insulin or vaccines (24 pens) at 2 to 8 °C. The payload sits in an aluminium liner wrapped in a phase-change material (PCM) that melts at 5 °C, inside 25 mm vacuum-insulated panels. A Peltier module freezes the PCM from a 12 V socket, a solar panel or a USB-C charger, and a small LiFePO4 battery keeps it running on the road. A two-phase loop thermosiphon connects the two and carries heat in one direction only, so a stopped Peltier does not leak heat back in. A logger records the payload temperature every minute and raises alarms. An aluminium cold plate under the lid PCM pack seats on the evaporator can when the lid closes, so the Peltier freezes all the PCM, and a second hardware cut-out on the liner stops a stuck-on driver from freezing the payload. The TRL 3 calculations (CPD-CAL-001 v0.2) give about 14.0 h with no power at 43 °C, 28.4 h off-grid at 32 °C and 17.5 h off-grid at 43 °C, for \$303 in parts and 5.53 kg empty. The hold times meet their targets with thin margins. The design misses two targets by small amounts: refreezing all the PCM takes 8.4 h (R8, 8 h) and the mass is 0.03 kg over (R12). Options for these are awaiting Amish. The parts cost is within the \$310 budget that Amish approved on 2026-09-26 (R15 met).

ColdPod

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Grey: Table top, phone for scale

Figure 1. ColdPod on a table, with a phone for scale. Cooling head on the right end, battery and electronics bay with the e-paper display on the left end. Rendered from the parametric model `cad/src/model.py`.

■ How it works

1. **Charge the cold.** At a clinic, in a vehicle or under a solar panel, the Peltier module runs at up to 25 W and freezes the PCM jacket through the thermosiphon and the aluminium evaporator can around it. The lid PCM pack sits on a 1.5 mm aluminium cold plate whose rim lands on the can rim when the lid closes, so it freezes through the can too. The same input charges the battery at about 12 W. A fully melted charge refreezes in about 8.4 h (the jacket in 6.9 h). The case works best if the payload goes in already cold from a refrigerator.
2. **Carry.** On the road, the controller runs the Peltier from the battery only as hard as needed to hold the evaporator can at about 2 °C, so the PCM stays frozen, and turns it off when the battery reaches 15 % so that the logger always has a reserve. A buck-boost driver feeds the module smooth DC from any input between 10 and 20 V; on/off PWM would cost about 60 % more power at 43 °C. The lid pack stays frozen on its plate.
3. **Hold passively.** When the battery is flat or the user switches cooling off, the PCM absorbs the heat that leaks through the walls; the plate lets the jacket and lid pack share the load. Because it melts at 5 °C, it cannot pull the payload below freezing the way water ice can.
4. **Watch.** A buffered probe among the payload, a liner probe, a cold-block probe and an ambient sensor are read every minute. The payload probe is logged with a timestamp on the controller's flash memory for 60 days or more.
5. **Alarm.** The display shows current, minimum and maximum temperature and time left. A buzzer, an LED and a phone notification warn after 10 min outside 2 to 8 °C, alarm at once at 0 °C or lower, and flag low battery, a failed sensor or a lid left open. The owner exports the log as a CSV file over Bluetooth Low Energy or USB. Nothing goes to the cloud.

ColdPod: heat flow, powered hold at 32 °C ambient (all values are estimates, CPD-CAL-001)

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Figure 2. Heat flow in powered hold at 32 °C ambient, can at 2 °C. All values are estimates from CPD-CAL-001: 0.092 W/K overall heat leak, TEC1-12703 class module at a coefficient of performance (COP) of 0.45, fan 0.6 W.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Proposed choice	Notes
1	Outer shell	3D-printed PETG or ASA tub, 3 mm walls, 261 x 196 x 164.5 mm	Protects the VIPs from knocks and punctures; walls cut from 4 mm to save mass
2	Lid and lid cold plate	Printed 5 mm cap with EPDM gasket and a 25 mm VIP plug; 1.5 mm aluminium cold plate, 205 x 140 mm, under the lid PCM pack	Latches closed; lid switch for the lid-open alarm; the plate rim lands on the evaporator can rim so the lid pack freezes and stays frozen
3	Bail handle and strap	Folding bail handle, padded shoulder strap	Rated 10 kg or more
4	Vacuum-insulated panels	Six panels, 25 mm, fumed silica core, ordered to size	VIPs cannot be cut or pierced; the heat-pipe crossing is a foam-filled gap between panels
5	PCM packs	About 0.97 kg of organic PCM melting at 5 °C (Rubitherm RT 5 HC class) in sealed HDPE pouches: a 0.69 kg jacket on four sides and the floor inside the evaporator can, and a 0.28 kg pack under the lid	Organic paraffins are stable for thousands of cycles (PATH PCM study)
6	Liner and rack	1.5 mm aluminium can, 175 x 110 x 80 mm outside (172 x 107 x 78.5 mm inside), with a removable printed rack for 24 pens in four layers of six	Spreads cold evenly around the payload
7	Evaporator can and thermosiphon	1.0 mm aluminium can lining the outer face of the PCM jacket (four sides and floor); two-phase loop thermosiphon: 8 mm copper evaporator loop around the base of the can, vapour riser and liquid return at the +X end, stainless sections through the wall, aluminium cold block 75 mm above the loop	One-way thermal link (design choices 2 and 3); sized in CPD-CAL-001, section G
8	Peltier module	40 x 40 mm, 127 couples, I _{max} about 3 A (TEC1-12703 class)	The TEC1-12706 cannot hold at 43 °C: its off-state conduction is too high (CPD-CAL-001, section E)
9	Heat sink and fan	Finned aluminium sink, 80 x 80 x 30 mm, 0.50 K/W or better with a 70 mm, 12 V fan and a finger guard	Rejects about 8.9 W at 32 °C and 19.7 W at 43 °C
10	End housings	Printed cooling head (+X end) and battery and electronics bay (−X end), 2 mm walls, vented	Keep the heat source and the battery outside the insulation
11	Battery	LiFePO ₄ , 4S1P 32700 cells, 12.8 V 6 Ah (76.8 Wh), BMS and fuse	Below the 100 Wh airline limit
12	Power board	USB-C PD sink (20 V), 12 V input (10 to 15 V) with reverse-polarity protection, LiFePO ₄ charger, buck-boost Peltier driver with an LC filter, fan driver; the cold-block and liner cut-outs wired in series in the Peltier supply	Off-the-shelf modules on perfboard; no custom PCB
13	Logger controller	nRF52840 module with real-time clock and 2 MB flash	Bluetooth Low Energy link to a phone
14			

#	Component	Proposed choice	Notes
	Temperature sensors and liner cut-out	Glycol-buffered payload probe, liner probe, cold-block probe, ambient sensor; bimetal cut-out on the liner opening at 3 °C	Digital sensors with ± 0.5 °C or better accuracy; the cut-out works without firmware
15	Display and alarm	2.13 in e-paper, buzzer, red and green LED	E-paper keeps the last reading visible with no power

Item 16 (wiring, fasteners, latches and consumables) is in the BOM but not modelled.

ColdPod: cutaway

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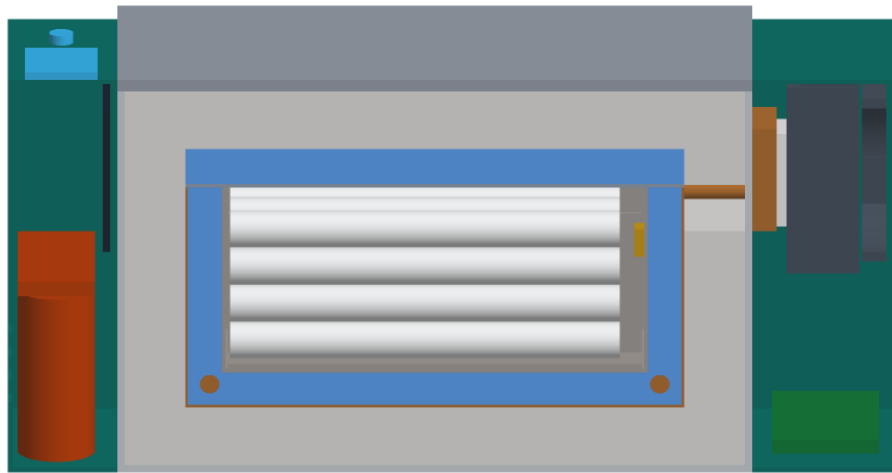


Figure 3. Section looking from the front. From the inside out: payload (white pens) in the rack and liner (6), PCM jacket and lid pack (5, blue) with the evaporator can and its loop (7, copper; the loop shows as two circles at the base), the lid cold plate (2) under the lid pack, VIPs (4, light grey) and shell (1). The pipes (7) cross the right wall to the cold block, Peltier (8) and heat sink (9) in the cooling head. The battery (11) and logger (13) are in the left bay.

■ Numbers at TRL 3

All values come from the calculation note CPD-CAL-001 (docs/04-calcs/01-sizing.md), which states every assumption; the tags in brackets point to lines of its script, docs/04-calcs/sizing.py . They are paper estimates. The main inputs are an aged VIP conductivity of 0.007 W/(m·K) with explicit joint and edge losses, 180 kJ/kg of

usable latent heat inside 2 to 8 °C for the PCM ([Rubitherm RT range](#)), a Peltier model derived from datasheet maxima (single-stage modules typically reach a COP of 0.3 to 0.7, [ThermalLM](#)), and 65.3 Wh of the 76.8 Wh battery available for cooling.

Table 2. Hold times, charging, size, mass and cost.

Quantity	Value	Basis	Requirement
Payload	1.36 L; 24 insulin pens in four layers of six, up to 172 mm long	Liner less rack and probe [A2, A3]	R3 met
Heat leak	0.092 W/K: 1.83 W at 25 °C, 2.47 W at 32 °C, 3.48 W at 43 °C	Langmuir shape factor, joints, foam strip, pipes, wires [B9, B10]	
PCM	0.972 kg, 175 kJ (48.6 Wh) usable	Jacket 0.691 kg, lid pack 0.282 kg [C2, C3]	
Passive hold, PCM only	26.6 h at 25 °C, 19.7 h at 32 °C, 14.0 h at 43 °C	Jacket and lid pack pooled through the lid cold plate [D1, D2]	R4 at risk (10.8 h at +30 % leak)
Input power to hold	4.2 W at 25 °C, 7.5 W at 32 °C, 18.5 W at 43 °C	TEC1-12703 class, can at 2 °C, buck-boost driver [E3]	R7 met
Battery hold, then PCM	15.5 + 26.6 = 42.1 h at 25 °C; 8.7 + 19.7 = 28.4 h at 32 °C ; 3.5 + 14.0 = 17.5 h at 43 °C	65.3 Wh, then the PCM [F2]	R5 and R6 at risk
PCM refreeze from melted	8.4 h for all the PCM at 25 °C (jacket 6.9 h, lid pack 8.4 h)	Time-stepped model, 25 W to the module [G4]	R8 not met (8 h)
Battery recharge	about 6.7 h at 12 W alongside the refreeze	45 W USB-C PD budget [G8]	
Heat rejected at the heat sink	5.3 W at 25 °C, 8.9 W at 32 °C, 19.8 W at 43 °C; sink base about 53 °C at 43 °C	Heat lifted plus module input [E6, E7]	Fan needed
Thermosiphon	Forward 2.50 W/K, reverse 0.0058 W/K; full function to 18° of tilt with the cooling head down	[H1, H5]	
Freeze fault	Liner settles near 2.0 °C after a stuck-on driver, with the 3 °C liner cut-out	Lumped estimate [I5]	R2 met
Logger reserve	about 96 days	15 % of 76.8 Wh at a 5 mW allowance [J4]	R11 met
Log storage	1.38 MB for 60 days	1 record per minute, 16 bytes [J1]	R9 met on 2 MB flash
Size	366 x 212 x 207 mm (14.4 x 8.3 x 8.1 in) overall	Handle up [A6]	R13 met
Mass, empty	5.53 kg (12.2 lb)	Model volumes and densities [K2]; thinner printed parts save 0.21 kg, the lid cold plate adds 0.12 kg	R12 not met (5.5 kg)
Parts cost	\$303	bom/bom.csv [L2]	R15 met (\$310)

■ Key design choices

Amish decided choices 1 to 9 on 2026-09-25, going with the recommendation in each case (CPD-DDR-001). The options are kept here for the record.

1. **PCM at 5 °C instead of water ice (decided, D1).** Water ice stores about 334 kJ/kg, nearly twice as much, but a frozen ice pack sits at 0 °C or below and is a known cause of vaccine freezing ([Hanson et al., 2017](#)). A 5 °C PCM keeps the whole payload space inside the window. Options were (a) 5 °C organic PCM; (b) water with a conditioning step; (c) a 3 °C PCM. Decision: (a).
2. **A thermosiphon between the Peltier and the PCM (decided, D2).** A Peltier module that is switched off conducts heat well: 0.27 W/K for the chosen module, about three times the whole box's heat leak. Bolted straight to the liner, it would cut the passive hold at 43 °C to about 3.6 h. A gravity thermosiphon, with its condenser above its evaporator, only carries heat upward and out, so it acts as a thermal diode (forward to reverse ratio about 430). Options were (a) thermosiphon; (b) a mechanical disconnect that lifts the cold block away when power is lost; (c) direct mounting. Decision: (a), with (b) as the fallback if the thermosiphon cannot be made reliable at this size. TRL 3 sizes it as a two-phase loop.
3. **Evaporator on the outer face of the PCM, not on the liner (decided, D3).** Keeps the coldest surface (down to −2 °C during refreezing) separated from the payload by the PCM, which supports R2. At TRL 3 this is sized as an aluminium can lining the outer face of the jacket, because a small plate cannot freeze a paraffin jacket that wraps the liner.
4. **LiFePO₄, 76.8 Wh (decided, D4).** LiFePO₄ tolerates heat better than other lithium-ion chemistries and 76.8 Wh stays under the 100 Wh airline limit. The rejected 4S2P option (153.6 Wh, about \$30 more and 0.6 kg heavier) would give about 18.6 h at 43 °C off-grid.
5. **R6 redefined (decided, D5).** Accept about 16 h at 43 °C off-grid and run from a vehicle 12 V socket or a solar panel on long hot trips, rather than a larger battery or 35 mm VIPs with 1.6 kg of PCM (about 400 mm long and over 6 kg). Revisit after field data on real trip temperatures.
6. **Vacuum-insulated panels instead of foam (decided, D7).** Polyurethane foam of the same thickness (about 0.024 W/(m·K)) gives 0.161 W/K rather than 0.092 W/K, so the passive hold at 43 °C would drop from 14.0 h to about 7.9 h (CPD-CAL-001, P1 and P2), but it would save about \$45 and some mass. VIPs are used, with a foam version documented as a low-cost variant.
7. **Local alarms and Bluetooth Low Energy only (decided, D8).** No cellular or LoRa radio in the first build (either would add \$20 to \$40 and more power).
8. **Alarm thresholds (decided, D9).** Warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; alarm if the liner probe reaches 1 °C (an early freeze warning); adjustable per product.
9. **Budget raised to \$300 (decided, D10).** Kept at \$300 by CPD-DDR-002; the parts cost after the TRL 3 changes is \$303.

Amish decided the TRL 3 review items on 2026-09-25, again going with the recommendation in each case (CPD-DDR-002):

1. **Lid cold plate (decided, DDR-002 item 2).** A 1.5 mm aluminium plate under the lid PCM pack, whose rim seats on the evaporator can rim when the lid closes (about \$5 and 0.12 kg). It gives the lid pack a cold path, so the Peltier refreezes all the PCM and the lid pack stays frozen in long powered holds. The other options were no lid pack with a 20 mm jacket, or conditioning the lid pack in a refrigerator. The gasket must still seal with the plate in place; that is a check for a built lid.

2. **Liner cut-out at 3 °C (decided, DDR-002 item 3).** A bimetal cut-out on the liner, in series with the −5 °C cold-block cut-out in the Peltier supply (about \$3). After a stuck-on driver fault the liner settles near 2.0 °C.
3. **Thinner printed parts (decided, DDR-002 item 4).** Shell 3 mm (was 4 mm), lid cap 5 mm (was 8 mm), end housings 2 mm (was 3 mm), saving about 0.21 kg. Confirming the mass by weighing is TRL 4 work, on hold.
4. **Evaporator can and loop thermosiphon confirmed (decided, DDR-002 item 5).** As sized in CPD-CAL-001: a 1.0 mm aluminium can around the jacket fed by an 8 mm copper loop. The mechanical disconnect (choice 2b) stays ready as the fallback if a maker cannot charge the loop.
5. **Peltier class and drive confirmed (decided, DDR-002 item 6).** TEC1-12703 class module, smooth DC through an LC filter, heat sink of 0.50 K/W or better with the fan.
6. **Buck-boost driver (decided, DDR-002 item 7).** The Peltier driver can raise the module voltage above a low vehicle input (10 V), rather than narrowing the 12 V input range to 11 to 15 V (about \$4 more).
7. **Budget kept at \$300 (decided, DDR-002 item 8), then raised to \$310 (decided by Amish, 2026-09-26, DDR-002 N3).** A VIP quote is to come before any purchase decision; purchasing is TRL 4 work, on hold.

ColdPod: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Outer shell
- 2 Lid cap with gasket
- 2 Lid cold plate, 1.5 mm aluminium
- 3 Bail handle and shoulder strap
- 4 Vacuum-insulated panels, 25 mm
- 5 PCM jacket and lid pack, 5 °C
- 6 Aluminium liner and payload rack
- 7 Evaporator can, loop thermosiphon, cold block
- 8 Peltier module, 40 x 40 mm
- 9 Hot-side heat sink and fan
- 10 End housings (cooling head, battery bay)
- 11 LiFePO4 pack, 12.8 V 6 Ah, with BMS
- 12 Power board (USB-C PD, 12 V, drivers)
- 13 Logger controller (nRF52840 class)
- 14 Temperature sensors and liner cut-out
- 15 E-paper display and alarm

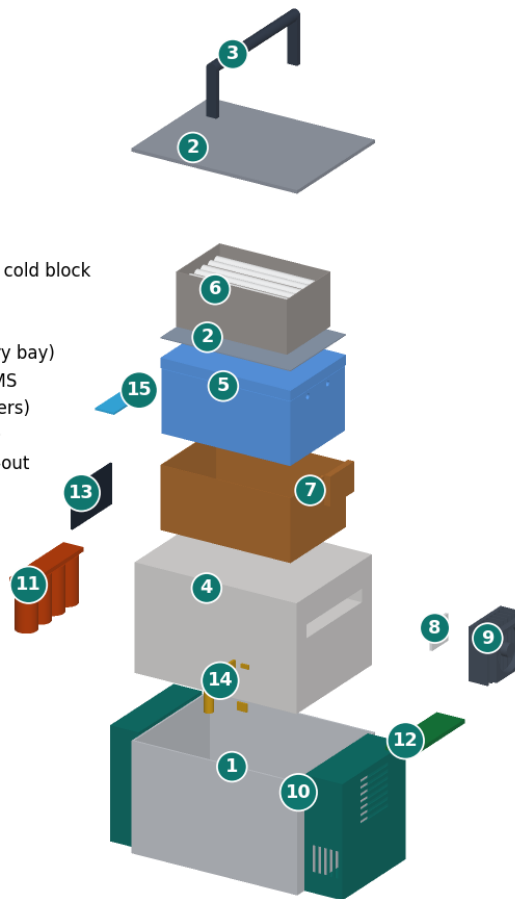


Figure 4. Exploded view with BOM numbers, from the parametric model. The payload pens rise with the liner (6); the evaporator can, loop and cold block (7) sit between the VIPs (4) and the PCM (5).

■ Safety

Safety: ColdPod is a research and educational prototype. It is not a medical device, is not WHO-prequalified and has not been cleared or approved by any regulator. Do not use it as the only protection for vaccines, insulin or other medicines. Follow national immunization program and manufacturer storage instructions, and use a qualified carrier and a calibrated logger for real products.

Safety: The LiFePO₄ pack stores 76.8 Wh. Use it only with its BMS and fuse, charge it only between 0 and 45 °C (the controller must block charging outside that range), and never charge a damaged, swollen or wet pack. The battery sits in its own vented bay outside the insulation, away from the Peltier's hot side.

Safety: The PCM is a paraffin, which is combustible. It stays sealed in HDPE pouches, separated from the battery bay by the VIPs and the shell. Replace any leaking pouch; paraffin can also soften some plastics ([PATH PCM study](#)).

- **Hot surfaces.** At 43 °C ambient the heat sink base reaches about 53 °C (estimate, CPD-CAL-001, E7). It is enclosed by the vented cooling head, and the vents must not be blocked, covered or placed against skin or a bag.
- **Moving parts.** The fan has a finger guard inside the cooling head grille.
- **Electrical.** All voltages are 20 V DC or less. There are no mains-voltage parts in the box. Inputs are fused and protected against reverse polarity.
- **Freezing fault.** A stuck-on Peltier driver is the main way to freeze the payload. Two hardware thermostats in series cut power to the Peltier independent of the firmware: one on the cold block at −5 °C and one on the liner at 3 °C. CPD-CAL-001 (section I) shows that the cold-block cut-out alone would let the liner follow the can down to about −4 °C; with the liner cut-out the liner settles near 2.0 °C. The liner and payload probes also raise alarms. The lid cold plate reaches −2 °C only while refreezing, which is done with the box empty; it touches no payload.
- **Condensation.** Water condenses on the cold block and pipes. The cooling head drains outward and the electronics are conformal-coated.
- **VIP damage.** A punctured VIP loses most of its insulation value without any visible change. The shell protects the panels, and a rising Peltier duty cycle should trigger a service warning.
- **Lifting.** About 6.2 kg when loaded; carry with the strap across the body.

■ Open questions after TRL 3

- Refreeze (R8): with the lid pack now frozen actively, all the PCM takes 8.4 h against 8 h. Options are in docs/REVIEW.md, awaiting Amish.
- Mass (R12): 5.53 kg against 5.5 kg after the thinner printed parts and the lid cold plate. Awaiting Amish; weighing a built case is TRL 4 work, on hold.
- Cost (R15): \$303 against the \$310 budget, raised from \$300 by Amish on 2026-09-26 (CPD-DDR-002 N3). Met; a VIP quote could still move it either way.
- Does the lid gasket still seal with the cold plate in place, and does the plate rim seat well enough on the can rim (assumed 0.5 K/W)?
- Can the loop thermosiphon be made and charged reliably at this size, and does a motorbike carrier stay within about 18° of tilt with the cooling head down? If not, the mechanical disconnect (choice 2b) is the fallback.

- Confirm the off-state conductance and real COP of the chosen module with the chosen heat sink, and the sink's resistance with the fan.
- Confirm the VIP supplier, panel sizes, aged conductivity and edge losses; joints and edges are about 45 % of the heat leak.
- Confirm the usable latent heat of the PCM between 2 and 8 °C, its supercooling and its behavior over many cycles.
- Where to place the payload probe so that it represents the warmest pen, not the average.
- Validate trip profiles, loads, alarm behavior and price with users through a partner (partner choice open, awaiting Amish).

Drawings and media: [general arrangement CPD-DWG-001](#), [concept blueprint sheet](#), [interactive 3D model](#).